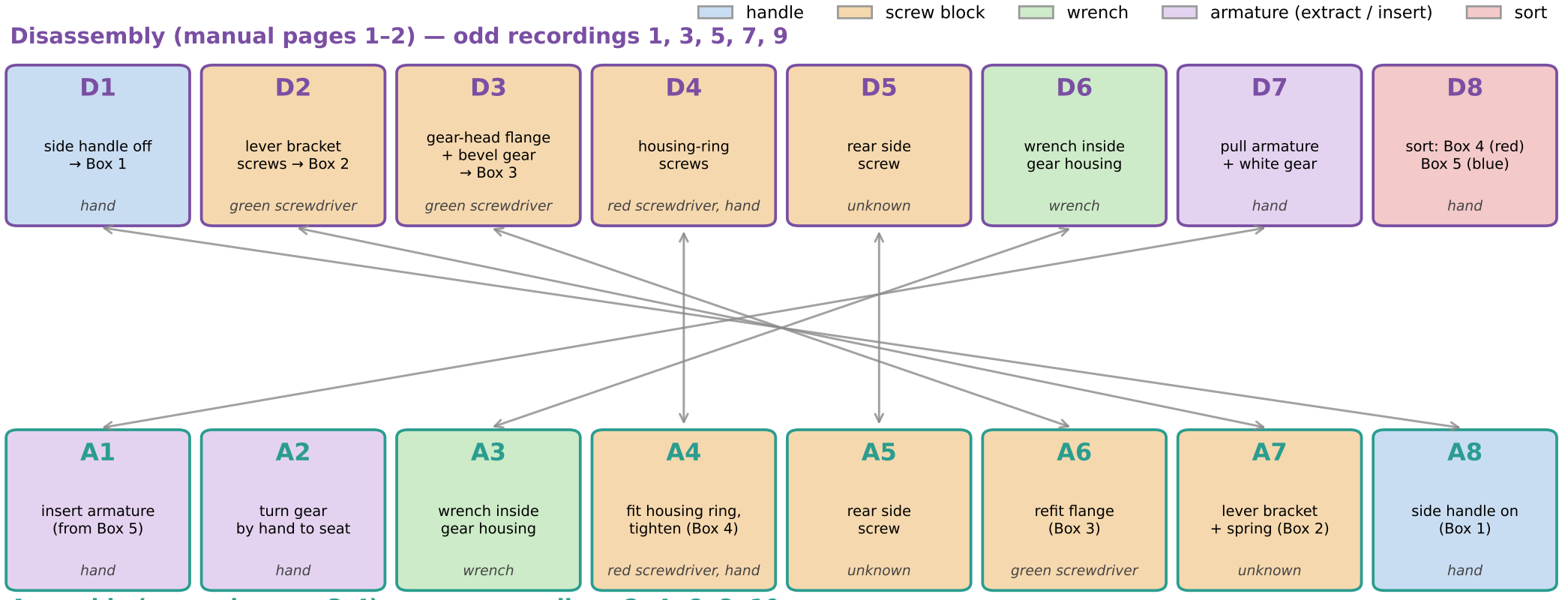
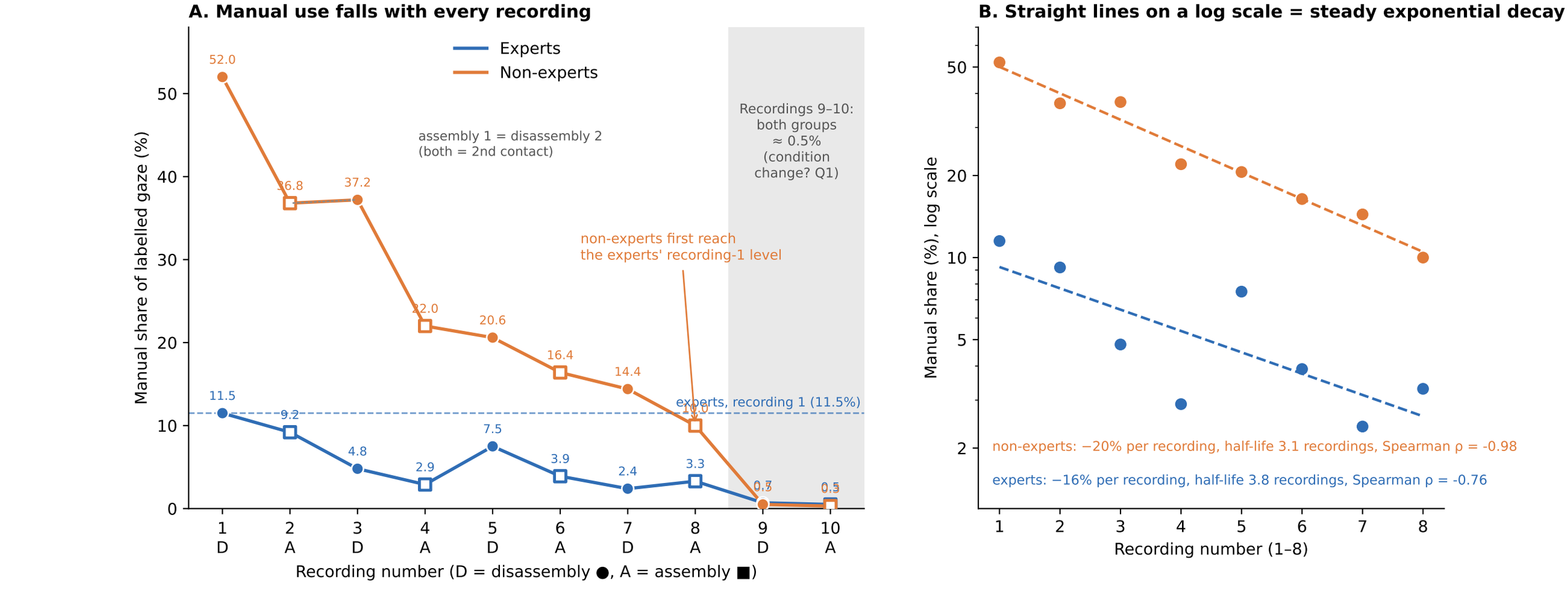
Figure 0. The manual as a process: assembly is disassembly reversed



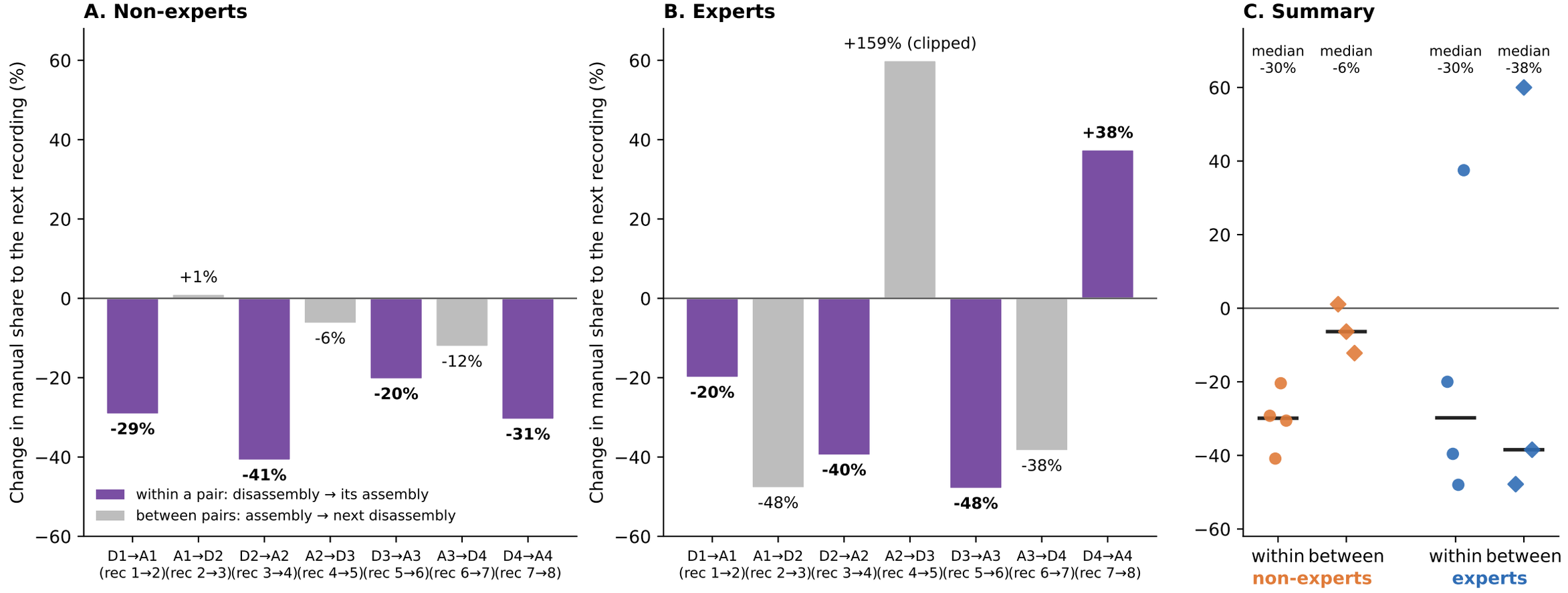
Source: manual_step_catalog.csv (visual reading of Anleitung Demontage 16.03.2025). Arrows join each assembly step to the disassembly step it reverses.

Figure 1. Manual use is a learning curve across the ten recordings



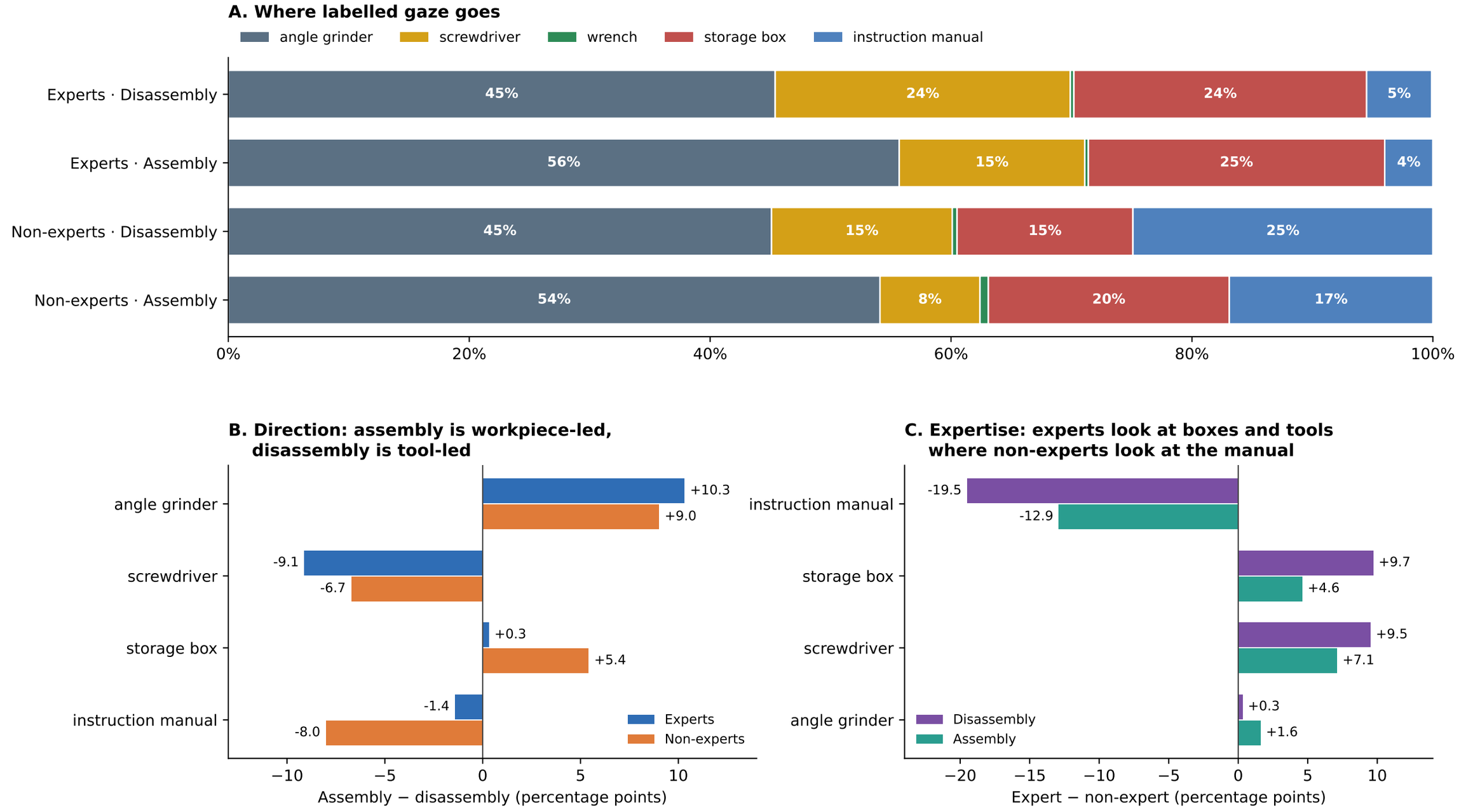
Source: manual_share_by_task_repetition.csv → chronological_manual_curve.csv. Participant means, 36 participants (17 experts, 19 non-experts), 360 recordings. Fit: log-linear least squares, recordings 1–8.

Figure 2. Non-experts drop the manual inside each disassembly → assembly pair, not between pairs



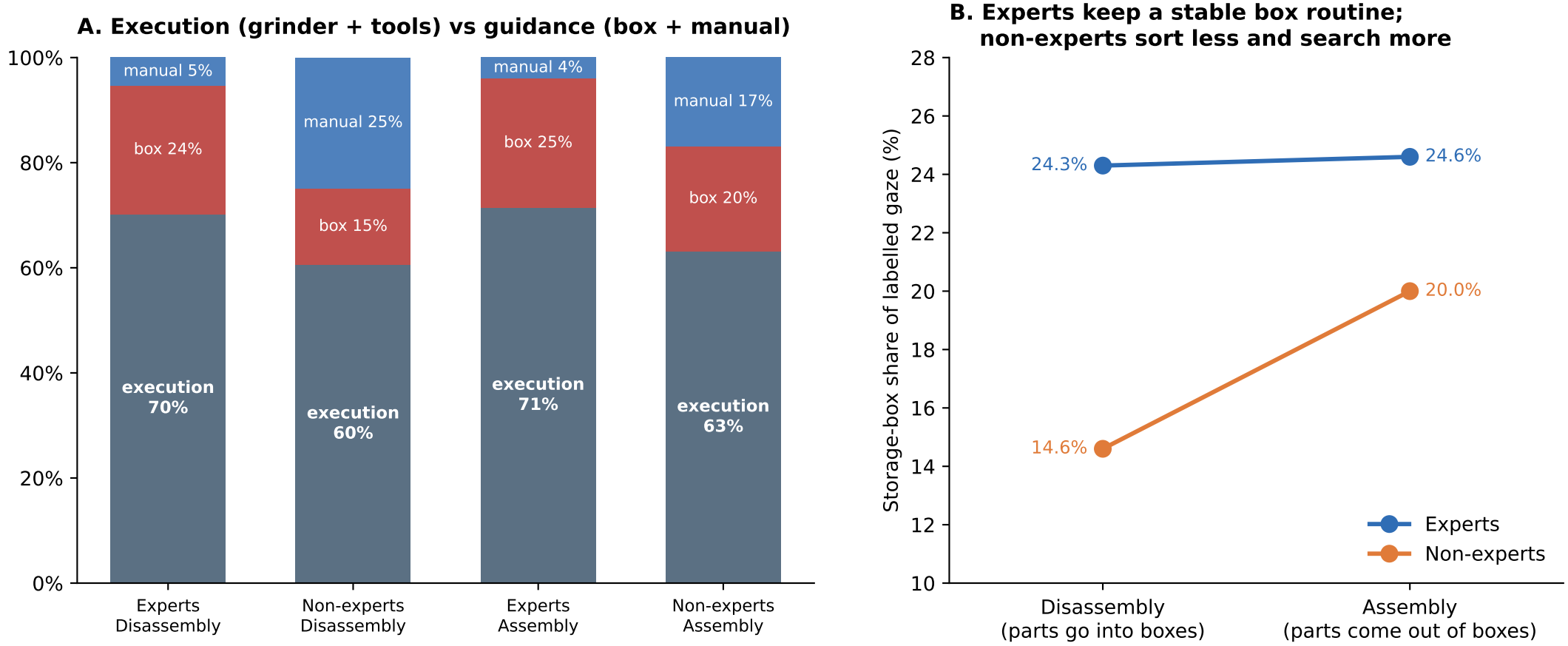
Source: manual_share_by_task_repetition.csv → manual_change_within_vs_between_pairs.csv (recordings 1–8). Change = (next − current) / current, participant means.

Figure 3. Attention signatures of direction and expertise



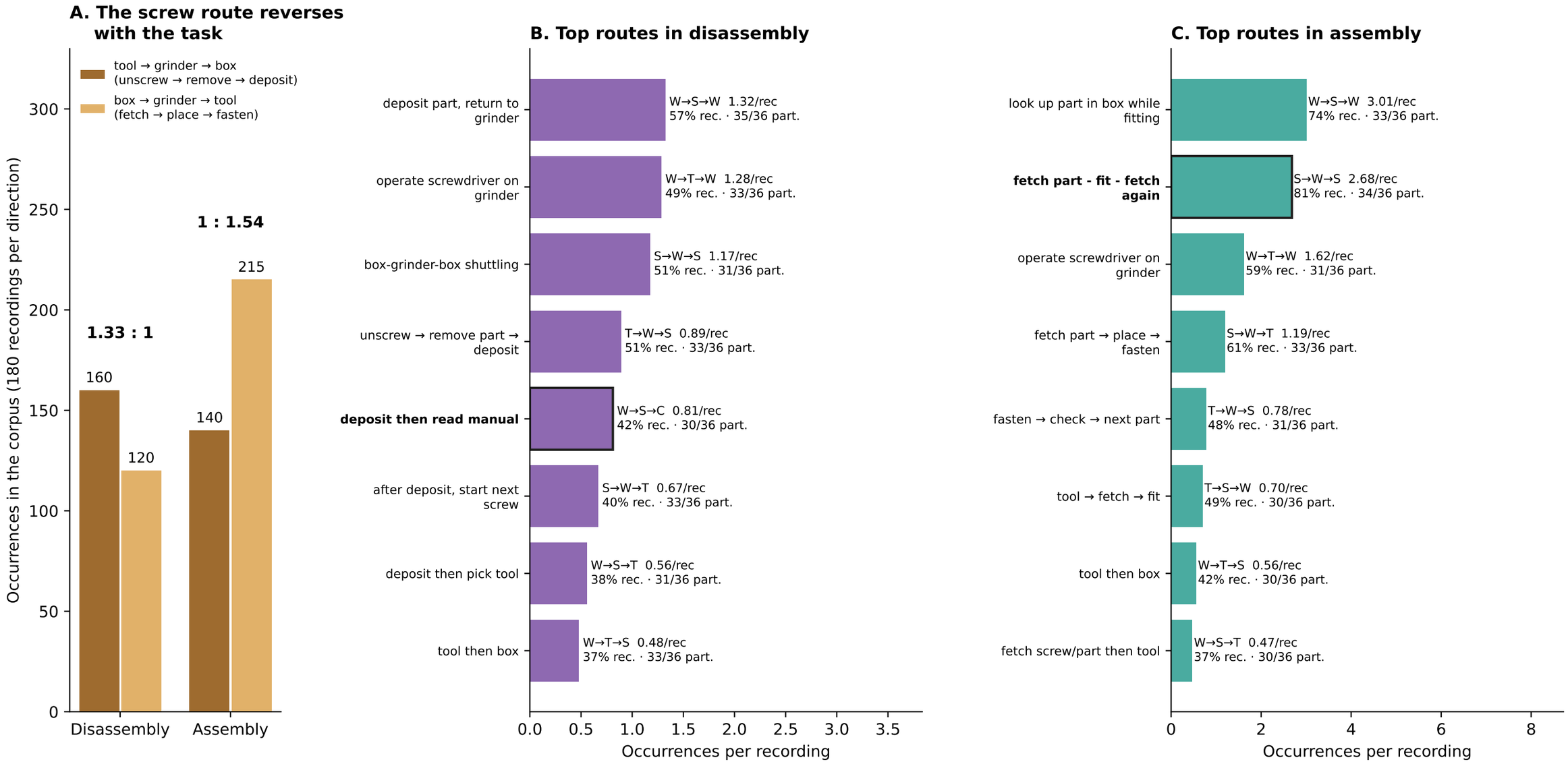
Source: attention_share_by_task_expertise.csv → direction_effect.csv, expertise_gap.csv. Share of labelled fixation time, participant means, 36 participants (17 experts, 19 non-experts), 360 recordings.

Figure 4. Experts do, non-experts read



Source: attention_budget.csv, attention_share_by_task_expertise.csv. Participant means, 36 participants (17 experts, 19 non-experts), 360 recordings.

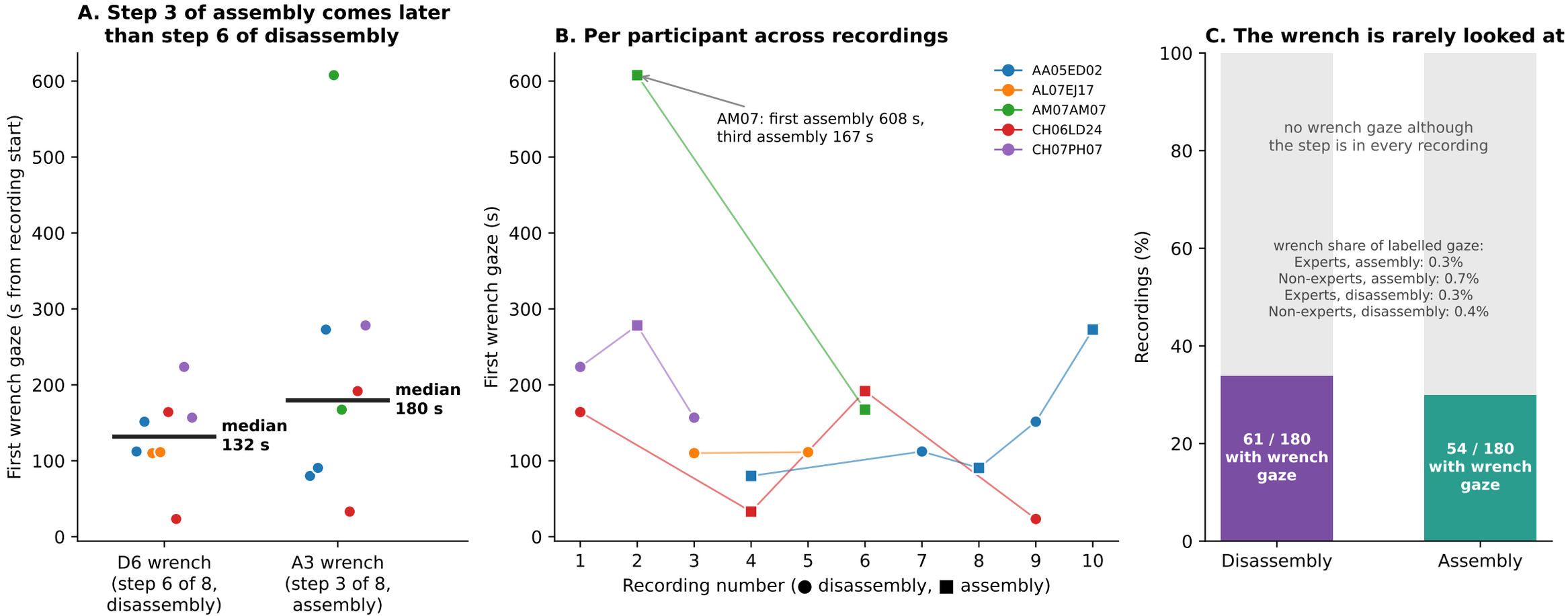
Figure 5. Gaze routes follow the manual's action order



W = grinder, S = storage box, T = tool, C = manual. Bold outline: deposit → read (step-boundary marker) and fetch → fit → fetch (assembly retrieval loop).

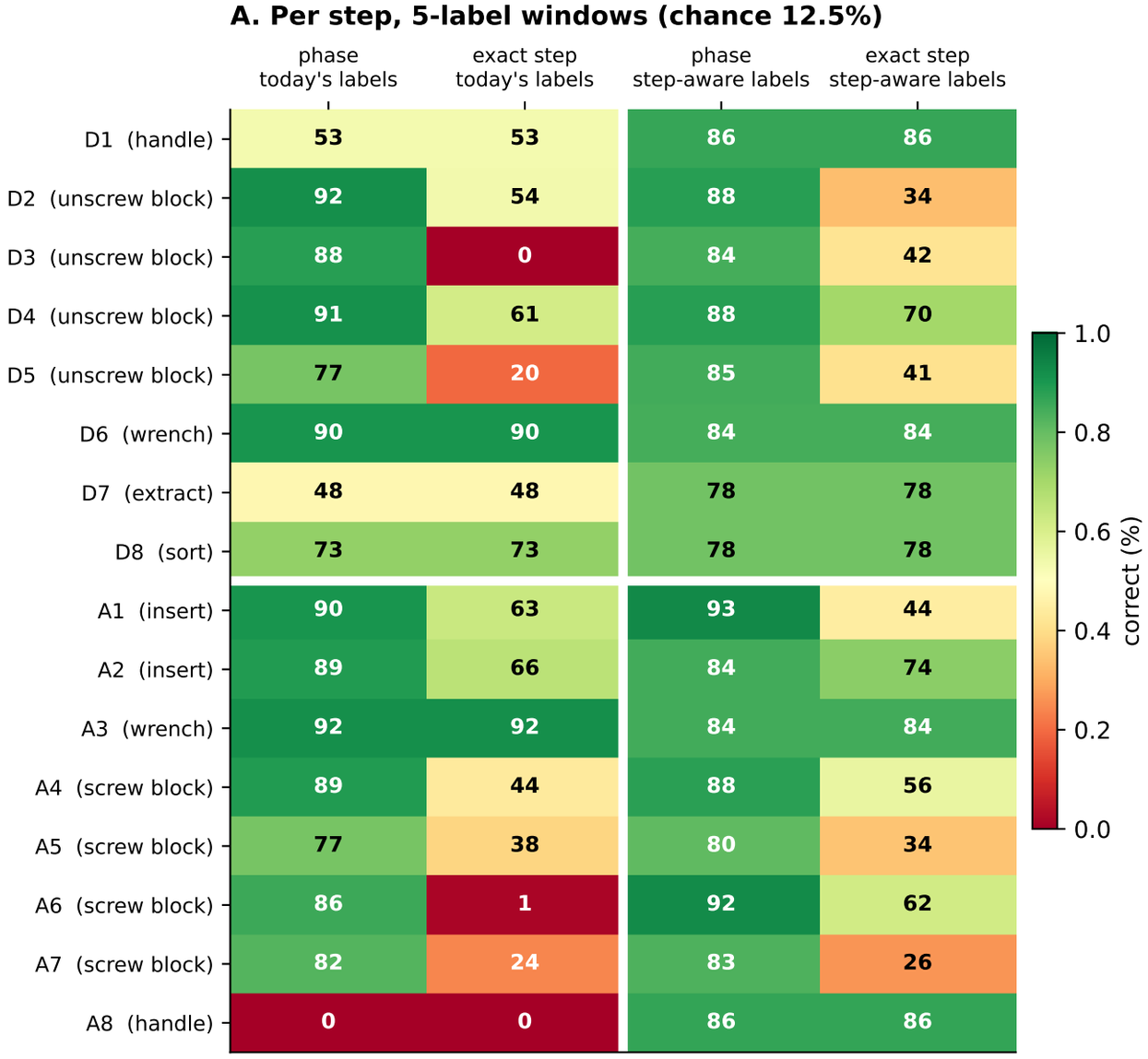
Source: top_routes_by_direction.csv → routes_mapped_to_manual.csv. Three-object routes; participants = number of the 36 who show the route at least once.

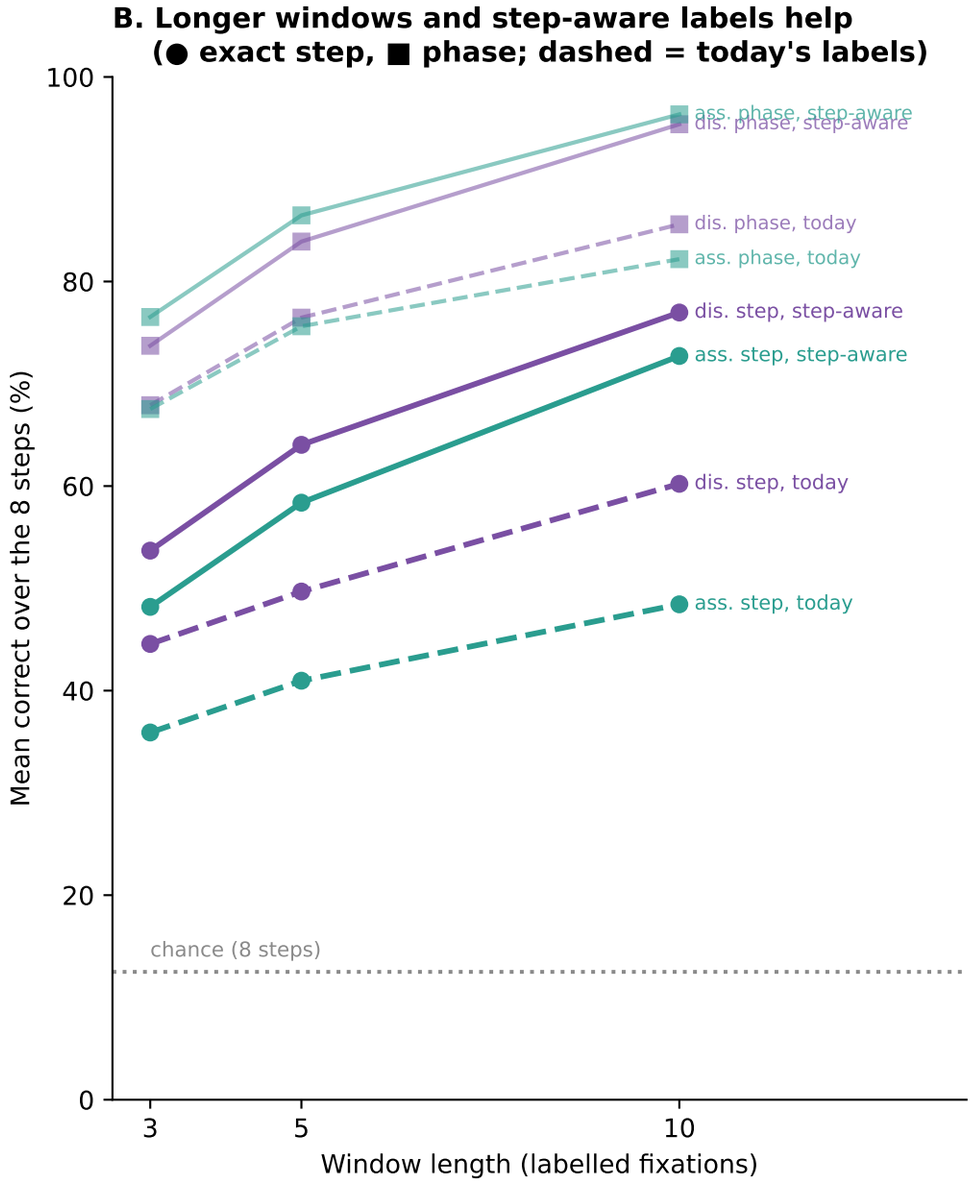
Figure 6. Timing anchor: the wrench step (D6 / A3)



Source: wrench_intervals_report_excerpt.csv → wrench_anchor_timing_excerpt.csv (16 recordings, 5 participants: the excerpt printed in the report); coverage 61/54 of 180 and shares from the report. Indicative, not a full-corpus estimate.

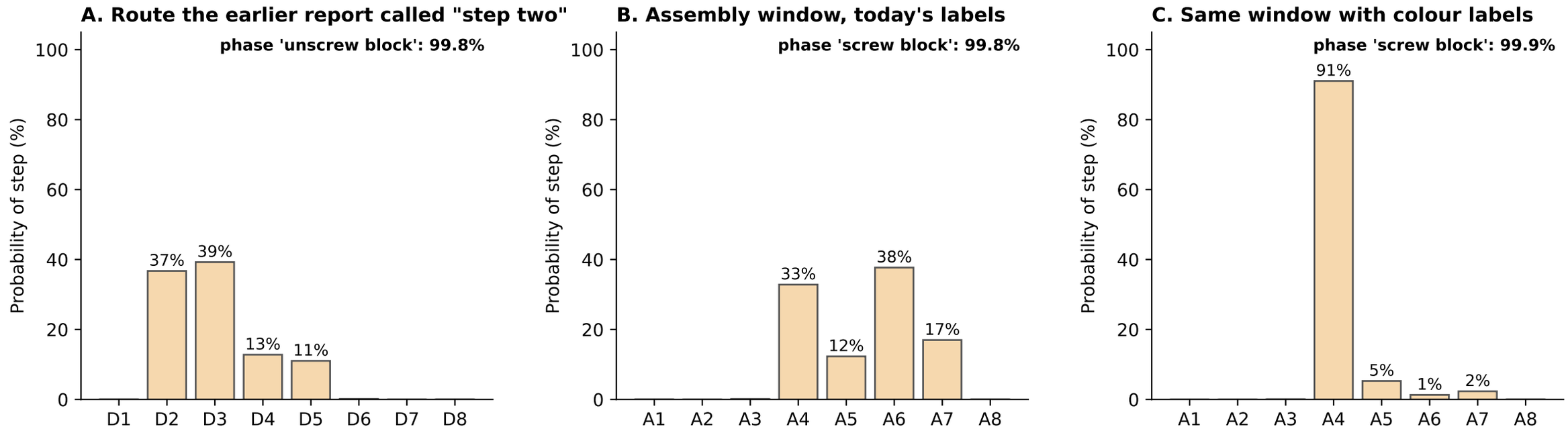
Figure 7. How well can a gaze window name the manual step?





Source: step_discriminability_model.csv — windows simulated from each step's expected gaze profile (configs/manual_steps.yaml), 400 per step. Shows what a label set can separate, not measured accuracy.

Figure 8. Predicting the manual step from a gaze sequence



Model: manual step profiles (configs/manual_steps.yaml), posterior over steps from the objects in the window. Sequence A is a real route from five_label_route_occurrences.csv (NG06NG25_07, KJ03JM25_10).